

SAFETY DATA SHEET

Hydrochloric Acid 22 Degree

Revision Date 10/7/2024

SECTION – 1 CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

Product Name Hydrochloric Acid 22 Degree **Item**

Product Use Acidification And pH Adjustment

Company Name Dakota Distributing **Office** (817) 453-7776
 2520 Chambers St. **Fax**
 Venus, TX 76084 **Web** dakotadistributing.com

EMERGENCY TELEPHONE NUMBER INFOTRAC (800) 535-5053

SECTION – 2 HAZARD IDENTIFICATION

Pictogram



Classification in accordance with (29 CFR 1910.1200)
US OSHA / HCS 2012 regulation

Signal Word Danger

Hazards PHYSICAL / HEALTH / ENVIRONMENTAL HAZARD STATEMENTS

May be corrosive to metals
 Harmful if swallowed
 Causes severe skin burns and eye damage
 Causes serious eye damage
 May cause respiratory irritation
 May cause long lasting harmful effects to aquatic life

HAZARD CATEGORY CLASSIFICATION CODE

Category 1	Corrosive to Metals	H290
Category 4	Acute Toxicity (Oral)	H302
Category 1B	Skin & Eye (Corrosion)	H314
Category 1	Eye (Damage / Irritation)	H318
Category 3	STOT Single Exposure	H335
Category 4	Chronic Toxicity (Aquatic)	H413

Precautions HANDLING / PROTECTION / FIRE / STORAGE / DISPOSAL

CODE

Keep only in original container	P234
Avoid breathing dust / fume / gas / mist / vapors / spray	P261
Do not get in eyes, on skin, or on clothing	P262
Wash thoroughly after handling	P264
Do not eat, drink or smoke when using this product	P270
Use only outdoors or in a well-ventilated area	P271
Avoid release to the environment	P273
Wear protective gloves / protective clothing / eye protection / face protection	P280
In case of inadequate ventilation wear respiratory protection	P285
Absorb spillage to prevent material damage	P390
Store in a well-ventilated place, Store locked up, Keep container tightly closed	P403+P405+P233
Store in corrosive resistant container	P406
Dispose of material in accordance with all State and Federal Guidelines and Regulations	P501

SECTION – 3 COMPOSITION INFORMATION

(Exact percentage of the listed chemicals of composition has been withheld as a trade secret)

<u>CHEMICAL NAME</u>	<u>COMMON NAME AND SYNONYMS</u>	<u>CAS #</u>	<u>IMPURITIES</u>	<u>PERCENT</u>
Hydrochloric Acid, 22° Baume	HCl, Muriatic Acid, Hydrogen Chloride Aqueous	7647-01-0	Water 63.05 to 74.00%	26 - 36.9%

SECTION – 4 FIRST AID MEASURES

Eye Contact Immediately flush eyes with cold water for at least 15 minutes while lifting upper and lower eyelids, Remove contact lenses if present and easy to do without injury to the eye and continue rinsing, Obtain immediate medical attention, preferably from an ophthalmologist or Emergency Room

Skin Contact Immediately wash contaminated skin with a nonabrasive soap and plenty of water for at least 15 minutes, Be sure to remove any contaminated clothing and wash before reuse, If irritation is present or occurs obtain medical attention

Inhaled Move person to fresh air, if they have problem breathing, show signs of overexposure or feel unwell obtain medical attention

Ingested DO NOT INDUCE VOMITING, unless directed to do so by medical personnel, If person is fully conscious, rinse mouth with water, Call a physician, or poison control center, and get medical attention, If vomiting occurs, keep head below hips to prevent aspiration into the lungs

Important Effects Material is destructive to, eyes, respiratory system, skin

Important Symptoms Symptoms may include, corrosive burns to eyes or skin, respiratory irritation

SECTION – 5 FIRE FIGHTING MEASURES

Extinguishing Media	Not flammable: Use extinguishing media for surrounding fire
Explosion Hazard	Not applicable
Hazardous Decomposition	Burning or thermal decomposition can produce, flammable, hydrogen chloride gas
Protective Equipment	Use MSHA/NIOSH approved self-contained breathing apparatus and full protective gear

SECTION – 6 ACCIDENTAL RELEASE MEASURES

Emergency Procedures	Warn personnel of spill, Stop spill or release only if it can be done safely, Keep unprotected personnel from entering the hazard area, Ventilate area
Personal Precautions	Follow all safety precautions, Wear Personal Protective Equipment, Do not walk through spill, Contaminated surfaces will be extremely slippery, Do not touch damaged containers or spilled material unless wearing appropriate protective equipment
Protective Equipment	Safety Glasses, Gloves, Chemical Apron, Rubber Boots
Containment	Use sand, absorbent socks or pads to prevent spill from spreading, Prevent spill from entering sewers and public waters
Clean Up Procedures	Neutralize spill with Soda Ash (Sodium Carbonate), Lime (Calcium Hydroxide), or Limestone (Calcium Carbonate) with large amounts of water, NOTE: Neutralization will create heat and carbon dioxide gas, Interior spills require ample ventilation
Disposal	Dispose of material in accordance with all State and Federal Guidelines and Regulations, Contact a licensed waste disposal contractor for proper disposal

SECTION – 7 HANDLING AND STORAGE

Handling	Do not get in eyes, on skin, or clothing, Avoid breathing mist, vapors or fumes, Use appropriate safety equipment, and adequate ventilation, Do not smoke, eat or drink while using, Wash thoroughly with soap and water after handling, Avoid release to the environment
Storage	Keep container closed when not in use, Store in a cool place away from incompatible materials, Store in corrosive resistant container
Incompatible Materials	Incompatible with, calcium phosphide, chlorosulfonic acid, ethylene diamine, alkalies, amines, acetic anhydride, ammonium hydroxide, bases, ethylene imine, oleum (fuming sulfuric acid), beta propiolactone, uranium phosphide, vinyl acetate, organic compounds, perchloric acid, reducing agents, strong bases, sodium hydroxide, sulfuric acid, most metals, This listing is not all-inclusive

SECTION – 8 EXPOSURE CONTROLS / PERSONAL PROTECTION**EXPOSURE LIMITS**

CHEMICAL NAME	ACGIH (TWA 8)	ACGIH (STEL)	OSHA (TWA 8)	OSHA (CEIL)	NIOSH (TWA 10)	NIOSH (STEL)	Significant Exposure
Hydrochloric Acid, 22° Baume		2 ppm		5 ppm (7 mg/m³)			ED,SD

PERSONAL PROTECTION

HMIS HAZARD RATINGS	
Health	3
Flammability	0
Reactivity	0
Personal Protection	H

Eyes	Wear safety glasses or goggles or face shield when handling / using this material
Hands	Wear chemical resistant impervious gloves when handling / using this material
Lungs	Wear a MSHA / NIOSH approved respirator at or above listed TLV's or if irritation is experienced
Body	"If Situation Requires" - Wear chemical resistant impervious protective clothing if exposure is considered to be likely when handling / using this material
Response	Access to a drench shower with eye wash station is a recommended safety precaution for handling / using this type of material
Ventilation	Ventilate to keep vapors of this material below the lowest ppm listed above, If over Threshold Limit Value use a MSHA / NIOSH approved respirator for organic vapor, supplied air or self-contained breathing apparatus

SECTION – 9 PHYSICAL AND CHEMICAL PROPERTIES

Flash Point	Not Flammable	Specific Gravity / Density	1.13 to 1.19
Flammable Limits (v)	Not Flammable	pH (± 0.3)	< 1
Auto-Ignition Temp.	Not Flammable	Viscosity (mm²/s / cSt)	2.3
Physical State	Liquid	Melting / Freeze Point	-24.44 to -52.78 °C (-12 °F to -63 °F)
Appearance	Clear to Slightly Yellow Liquid	Boiling Point	110 °C (230 °F)
Odor	Sharp Pungent Acidic Odor	Vapor Density (air=1)	9.48 to 9.61
Odor Threshold	0.25 to 10 ppm	Vapor Pressure (mmHg)	50 to 60 mm
Solubility	100% (miscible)	Evaporation Rate (nBuAc=1)	ND
Volatiles	65%	Partition Coefficient	ND
VOC	0%	Molecular Weight (g/mol)	36.46
LVP-VOC	0%	Decomposition Temperature	ND

SECTION – 10 STABILITY AND REACTIVITY

Reactivity	No specific test data related to reactivity available for this product or its ingredients
Chemical Stability	Stable under normal ambient and anticipated conditions of use
Hazardous Polymerization	Will not occur
Conditions To Avoid	Incompatible materials
Incompatible Materials	Incompatible with, calcium phosphide, chlorosulfonic acid, ethylene diamine, alkalies, amines, acetic anhydride, ammonium hydroxide, bases, ethylene imine, oleum (fuming sulfuric acid), beta propiolactone, uranium phosphide, vinyl acetate, organic compounds, perchloric acid, reducing agents, strong bases, sodium hydroxide, sulfuric acid, most metals, This listing is not all-inclusive
Hazardous Decomposition	Burning or thermal decomposition can produce, flammable, hydrogen chloride gas

SECTION – 11 TOXICOLOGICAL INFORMATION**ROUTES OF EXPOSURE**

Eyes (Yes), Skin (Yes), Ingestion (Yes), Inhalation (Yes)

ACUTE SYMPTOMS OF SINGLE OVEREXPOSURE

Eyes	Causes serious eye damage, severe pain, corrosive burns
Skin	Can cause serious skin damage
Inhalation	Mist, vapor or fumes may cause, respiratory or mucosal irritations
Ingestion	Harmful if swallowed, Can cause corrosive burns, of the mouth, throat, esophagus, and stomach

CHRONIC SYMPTOMS OF PROLONGED OR REPEATED OVEREXPOSURE

Eyes	Causes serious eye damage, severe pain, corrosive burns, vision impairment
Skin	Causes serious skin damage, inflammation, ulcerations, corrosive burns
Inhalation	Mist, vapor or fumes can cause, respiratory or mucosal irritations, coughing, breathing difficulties, shortness of breath, inflammation and edema of the bronchi, pneumonitis, pulmonary edema
Ingestion	Harmful if swallowed, Can cause corrosive burns to digestive system tissue, Symptoms may include, nausea, vomiting, internal bleeding, May cause damage to organs

Acute Tox Calculated	Oral: 900 mg/kg	Dermal: 5,010 mg/kg	Inhaled: 162.5 mg/l
Acute Tox Category	Category 4 (Oral >300, ≤2,000 mg/kg), Not applicable (Dermal >2,000 mg/kg), Not applicable (Inhaled >20 mg/l) Vapors		
Target Organs	Skin, Eyes, Respiratory System		
Medical Conditions	Preexisting, eye, skin, respiratory, disorders may be aggravated by exposure to this product		
Notes to Physician	Treat symptoms		

CARCINOGENIC – This product contains concentrations above 0.1% of the following:

<u>CHEMICAL NAME</u>	<u>NTP</u>	<u>ACGIH</u>	<u>IARC</u>	<u>GHS Category</u>
None Listed	NA	NA	NA	NA

MUTAGENIC AND REPRODUCTIVE EFFECTS – This product contains concentrations above 0.1% of the following:

<u>CHEMICAL NAME</u>	<u>Germ Cell Mutagenicity</u>	<u>Toxic to Reproduction</u>
None Listed	NA	NA

COMPONENTS ACUTE TOXICITY

<u>CHEMICAL NAME</u>	<u>Type</u>	<u>Form</u>	<u>Subject</u>	<u>Result Value</u>	<u>Exposure Time</u>	<u>GHS Category</u>
Hydrochloric Acid	LD50	Oral	Rabbit	900 mg/kg		4 (>300, ≤2000 mg/kg)
	LCLo	Inhaled	Human	162.50 mg/l	4 Hours (Vapor)	(>20 mg/l)
	LD50	Dermal	Rabbit	5010 mg/kg		(>2000 mg/kg)
	LC50	Inhaled	Mouse	413 mg/l	4 Hours (Vapor)	(>20 mg/l)

SECTION – 12 ECOLOGICAL INFORMATION

<u>CHEMICAL NAME</u>	<u>Type</u>	<u>Subject</u>	<u>Subject Latin</u>	<u>Result Value</u>	<u>Exposure Time</u>	<u>GHS Category</u>
Hydrochloric Acid	LC50	Mosquitofish	(Gambusia affinis)	282 mg/l	96 Hours	4 (>100 mg/l)
	LC50	Golden Orfe	(Leuciscus Idus)	862 mg/l	96 Hours	4 (>100 mg/l)
Presistence And Degradability	Not applicable for inorganic substances					
Bioaccumulative Potential	There is no evidence to suggest bioaccumulation will occur					
Mobility In Soil	Soluble in water and is leachable through normal soil					
Other Adverse Effects	Harmful to aquatic organisms due to pH shift					

SECTION – 13 DISPOSAL CONSIDERATIONS

Disposal Statement	DO NOT DUMP INTO ANY SEWERS, ON THE GROUND, OR INTO ANY BODY OF WATER Dispose of any waste in accordance with all State and Federal Guidelines and Regulations
Container Disposal	Empty containers retain product residue (vapors, liquid or solid) observe all precautions when handling, Empty drums should be returned to distributor or taken to an approved waste handling site for recycling or disposal
Material Disposal	This material as supplied, when discarded or disposed of, is a hazardous waste according to Federal Regulations (40 CFR 261) due to its composition containing in some or all of its components, Under RCRA rules, it is the responsibility of the user of the product to determine, at the time of disposal, whether the material is a hazardous waste, Chemical additions, processing or otherwise altering this material may make the waste management information presented in this SDS incomplete, inaccurate, or otherwise inappropriate

SECTION – 14 TRANSPORT INFORMATION**DOT CLASSIFICATION**

UN Number	Proper Shipping Name n.o.s. (Chemicals) or "Limits"					
UN 1789	Hydrochloric Acid					
Hazard Class	Packing Group	Label Codes	Reportable Quantity (lb)	Response	Marine Pollutant	Hazard Label Secondary
8	II	Corrosive Liquid	(5,000)	157	No	
Additional Info:						

SECTION – 15 REGULATORY INFORMATION**TSCA**

CHEMICAL NAME	Sec 8(b) Active Inventory	Sec 8(d) Health And Safety	Sec 4(a) Chemical Test Rules	Sec 12(b) Export Notification
Hydrochloric Acid	Active			

REPORTABLE QUANTITIES

	Extremely Hazardous		Reportable Quantity	Emission Reporting		
CHEMICAL NAME	EPCRA TPQ Sec 302	EPCRA RQ Sec 304	CERCLA RQ Sec 103	TRI Sec 313	RCRA Code	RMP TQ Sec 112r
Hydrochloric Acid			5,000			

SARA

	Section 311		Section 311 / 312 Hazards			
CHEMICAL NAME	Hazardous Chemical	Acute	Chronic	Flammable	Pressure	Reactive
Hydrochloric Acid	Yes	Yes				

RIGHT TO KNOW

	STATE											
CHEMICAL NAME	CA	CT	FL	IL	LA	NJ	NY	PA	MI	MN	MA	RI WI
Hydrochloric Acid	Yes					Yes		Yes			Yes	Yes

CALIFORNIA

WARNING: This Product can expose you to chemicals (Listed below) known to the State of California to cause cancer, birth defects or reproductive harm. For more information go to www.P65Warnings.ca.gov

CHEMICAL NAME	CAS #	Birth Defects	Reproductive Harm	Carcinogen	Developmental
None Listed					

CLEAN AIR WATER ACTS

		Clean Air Acts			Clean Water Acts		
CHEMICAL NAME	CAS #	HAP	Ozone Class 1	Ozone Class 2	HS	PP	TP
None Listed							

INTERNATIONAL REGULATIONS – The components of this product are listed on the chemical inventories of the following countries:

CHEMICAL NAME	Australia	Canada	Europe (EINECS)	Japan	Korea	UK
Hydrochloric Acid	Yes	Yes	Yes	Yes	Yes	Yes

SECTION – 16 OTHER INFORMATION**SDS** **LEGEND DESCRIPTION**

~	Approximately	LC0	A concentration that is "Not" lethal to a given species in a given time
ACGIH	American Conference of Governmental Industrial Hygienists	LC50	A concentration that is lethal to 50% of a given species in a given time
CAS	Chemical Abstracts Service Registry	LD0	Dose that is "Not" lethal to a given species by a given route of exposure
CEIL	Ceiling Limit (15 minutes)	LD50	Dose that is lethal to 50% of a given species by a given route of exposure
CERCL	Comprehensive Environmental Response, Compensation, and Liability Act	LEL	Lower Explosive Limit
CI	Cochlear Impairment	NA	Not Applicable
CNS	Central Nervous System	ND	Not Determined
EC50	Concentration of a chemical that gives half-maximal response	NE	Not Established
EPA	Environmental Protection Agency	NFPA	National Fire Protection Association
Eye	(EI = Irritation) (ED = Damage) (EV = Visual Impairment)	NIOSH	National Institute for Occupational Safety and Health
GHS	Globally Harmonized System	NTP	National Toxicology Program
HAP	California Hazardous Air Pollutant Clean Air Act	OSHA	Occupational Safety and Health Administration
HMIS-A	Safety glasses	PEL	Permissible Exposure Limit (OSHA)
HMIS-B	Safety glasses, gloves	PNS	Peripheral Nervous System
HMIS-C	Safety glasses, gloves, chemical apron	PP	California Priority Pollutant under the Clean Water Act
HMIS-D	Face shield, gloves, chemical apron	REL	Recommended exposure limit (NIOSH)
HMIS-E	Safety glasses, gloves, dust respirator	RT / RS	Respiratory Tract / Respiratory System

HMIS-F Safety glasses, gloves, chemical apron, dust respirator
HMIS-G Safety glasses, gloves, vapor respirator
HMIS-H Splash goggles, gloves, chemical apron, vapor respirator
HMIS-I Safety glasses, gloves, dust and vapor respirator
HMIS-J Splash goggles, gloves, chemical apron, dust and vapor respirator
HMIS-K Air line hood or mask, gloves, full chemical suit, boots
HMIS-X Ask Supervisor
HS California Hazardous Substance under the Clean Water Act
IDHL Immediately dangerous to health and life
IG / IH (IG = Ingested) / (IH = Inhaled - Vapors / Mists / Gas)

Skin (SI = Irritation) (SD = Damage) (SA = Absorption) (SS = Sensitizer)
SARA Superfund Amendments and Reauthorization Act
STEL Short Term Exposure Limit (15 minutes)
TC Lo Lowest concentration that is toxic to a given species in a given time
TD Lo Lowest dose that is toxic to a given species
TLV Threshold Limit Value (ACGIH)
TP California Toxic Pollutant under the Clean Water Act
TSCA Toxic Substances Control Act
TWA Time Weighted Average (8 hours) - NOISH (10 hours)
UEL Upper Explosive Limit

Dakota Distributing

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-- End of Safety Data Sheet --

Supersedes Safety Data Sheet Dated